Question 4 [10 marks]

a) A test has been devised to measure a student's level of motivation during 1st year in university. The motivation scores on this test are approximately normally distributed with mean of 25 and a standard deviation of 6. The higher the score the greater the motivation to do well in university.

i) What percentage of students taking this test will have scores below 10? (1 mark)

ii) What percentage of students taking this test will have scores between 16 and 31? (3 marks)

iii) Aishah is told that 33% of the students taking the test have higher motivation scores that he does? What was Aishah's score? (3 marks)

b) A package delivery service divides their packages into weight classes. Suppose that packages in the weight class of 14 to 20 kg are uniformly distributed, meaning that all weights within that class are equally likely to occur. Let X denote the weight of a package.

i) Write the density function, $f(x)$ of X. (1 mark)

ii) Find the probability that a package weighs at least 15 pounds. (2 marks)

QUESTION 4 (14 MARKS)

The level of a particular pollutant, nitrogen oxide, in the exhaust of a hypothetical model of car, the Polluter, when driven in city traffic has approximately a normal distribution with a mean level of 2.1 grams per mile (g/m) and a standard deviation of 0.3 g/m.

a) What is the probability of selecting a Polluter car randomly and finding its nitrogen oxide emissions within the range of 2.0 g/m to 2.5 g/m? [3 marks]

b) If the Environmental Department mandates that a nitrogen oxide level of 2.7g/m. What proportion of Polluters would be in violation of the mandate? [3 marks]

c) What is the score corresponds to the top 10% of the nitrogen oxide emissions? [4 marks]

d) The company producing the Polluter must reduce the nitrogen oxide level so that at most 5% of its cars exceed the level of 2.7 g/m. If the standard deviation remains 0.3 g/m, to what value must the mean level be reduced so that at most 5% of Polluters